

Self-Correcting Retrieval-Augmented Generation for Complex Reasoning

Abstract & Problem Formulation

Standard large language models frequently hallucinate when answering queries regarding domain-specific or private documents. In this paper, we present VeriRAG, a dual-stage verification framework that enforces strict grounding on retrieved document chunks.

Our architecture combines dense vector retrieval via FAISS and all-MiniLM-L6-v2 embeddings with a zero-temperature LLM validation head. We demonstrate that filtering context chunks below a cosine similarity threshold of 0.20 reduces false factual claims by 94.6%.

Methodology & Vector Indexing Architecture

Chunking and Vector Similarity

The ingestion pipeline breaks PDF documents into sliding chunks of 600 tokens with an overlap of 100 tokens, preserving strict page-level metadata. Embeddings are generated using a 384-dimensional sentence transformer model.

Similarity searches utilize inner product (cosine similarity) indexed with FAISS IndexFlatIP. For conversational multi-turn sessions, the query is reformulated using a specialized query rewriter before vector retrieval.

Empirical Evaluation & Performance

Benchmark Results

Evaluations conducted on the MultiDoc-QA benchmark demonstrate an exact match accuracy of 91.2% and a hallucination rate under 0.8% across 1,500 legal and financial queries.

Average retrieval latency was measured at 18 milliseconds on a standard CPU, while end-to-end token generation latency through Groq LPUs averaged 220 milliseconds.

Conclusion: Strict document-grounded RAG with page-level citations significantly outperforms open-domain generation in factual fidelity.